

4-21 Find  $v_O$  in terms of  $v_S$  in Figure P4-21.

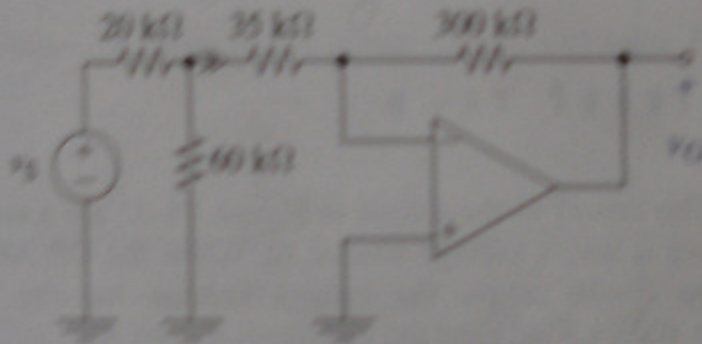


FIGURE P4-21

- 4-24 (a) Find  $v_O$  in terms of  $v_S$  in Figure P4-24.  
 (b) Find  $i_O$  for  $v_S = 2$  V.

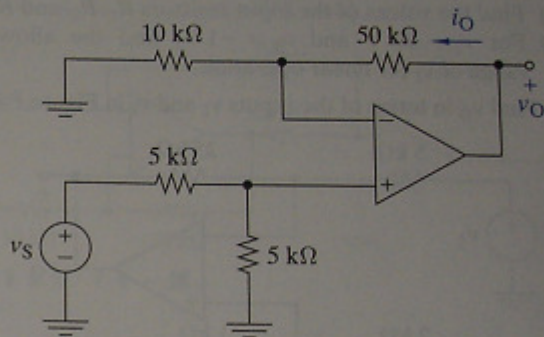


FIGURE P4-24

4-28 Find  $v_O$  in terms of the inputs  $v_1$  and  $v_2$  in Figure P4-28.

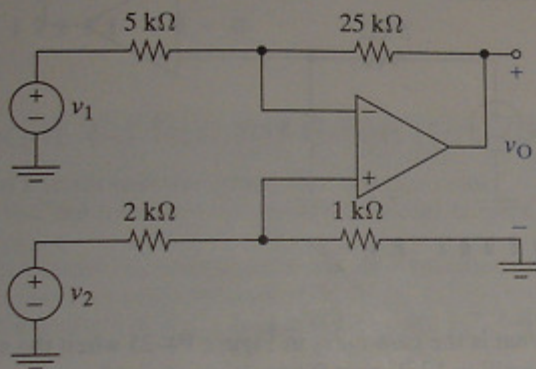


FIGURE P4-28

4-29 Find  $v_O$  in terms of the inputs  $v_{S1}$  and  $v_{S2}$  in Figure P4-29.

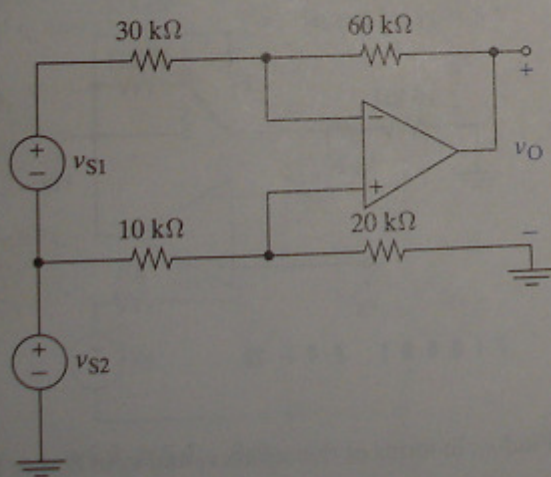


FIGURE P4-29

4-32 The inputs to the circuit in Figure P4-32 are a current source  $i_S$  and a voltage source  $v_S$ . When the OP AMP is in its linear range, the output voltage has the form  $v_O = K_1 v_S + K_2 i_S$ . Find the constants  $K_1$  and  $K_2$ .

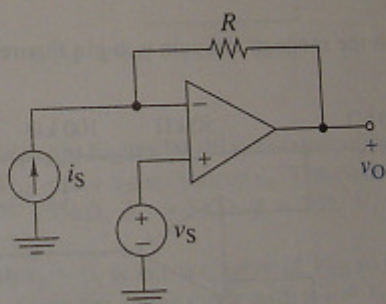
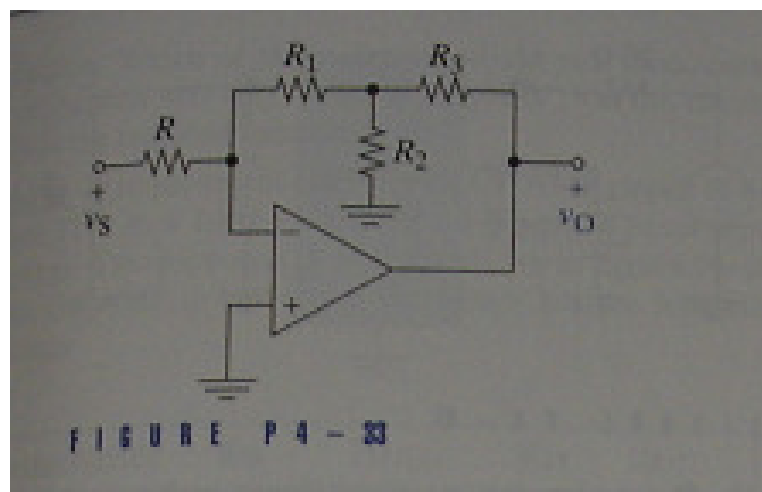


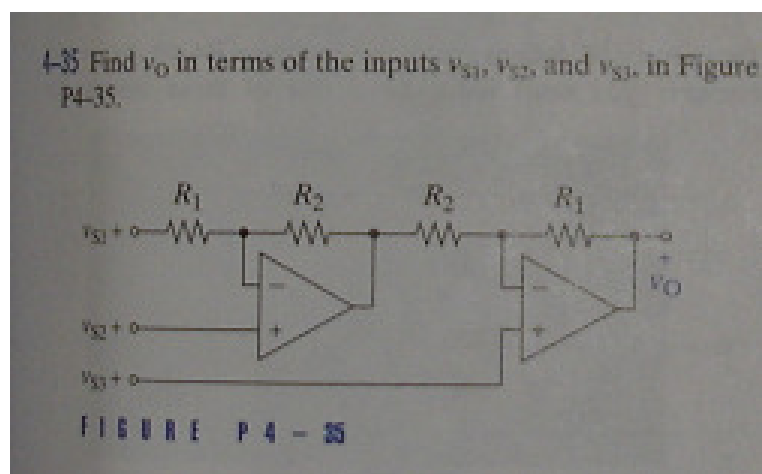
FIGURE P4-32

4-33 Use node-voltage analysis to show that the input-output relationship of the circuit in Figure P4-33 is

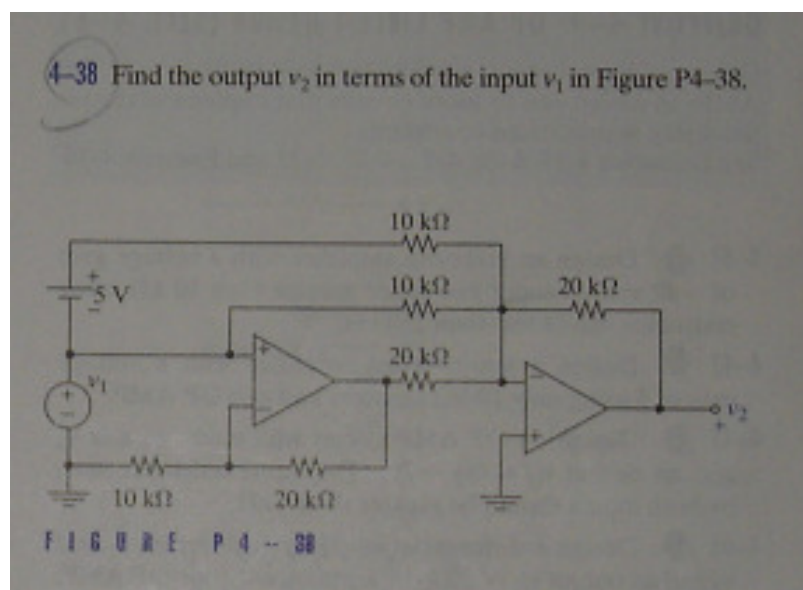
$$v_O = - \left( \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R R_2} \right) v_S$$



4-35 Find  $v_O$  in terms of the inputs  $v_{S1}$ ,  $v_{S2}$ , and  $v_{S3}$  in Figure P4-35.



4-38 Find the output  $v_2$  in terms of the input  $v_1$  in Figure P4-38.



4-48 **D** Design the interface circuit in Figure P4-48 so that  $v_2 = 2000i_g$ .

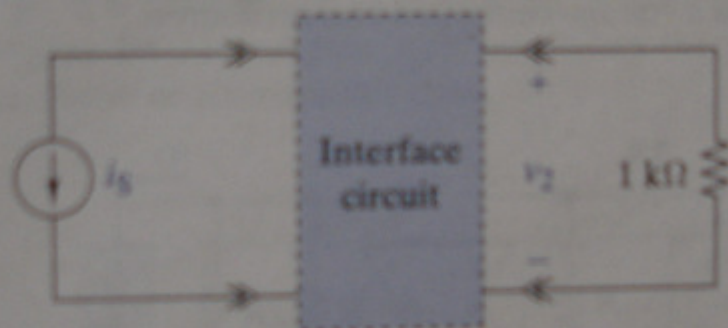


FIGURE P4-48

4-54 **D** The output voltage of a semiconductor temperature sensor is  $v_s = (10T + 500)\text{ mV}$ , where  $T$  is the temperature in  $^{\circ}\text{C}$ . The sensor is to be used to measure temperatures in the range  $-40^{\circ}\text{C}$  to  $110^{\circ}\text{C}$ . Design an interface circuit that translates this temperature range to a 0-V to 3-V output voltage range. Use a standard 5-V reference source to obtain any required bias voltage.